

The Kobon triangle problem for 18 lines: an upper bound of 94 for all arrangements, and a reduction of $K(18) = 93$ to a single local configuration

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Preprint. Computer-assisted proofs; no human refereeing or formal verification yet.

Abstract

The Kobon triangle problem asks for the largest number $K(n)$ of bounded triangular faces in an arrangement of n lines in the plane. For $n = 18$ the best known arrangements have 93 triangles.

- For *simple* arrangements (no three lines through a point) Blanc proved 93 is optimal.
- For arrangements with multiple points, the best bounds previously known to us are Tamura’s 96 and an unpublished 95.

We prove, with computer assistance, that **every arrangement of 18 lines or pseudo-lines, of any multiplicity, has at most 94 bounded triangles**. We also prove that an arrangement with 94 triangles cannot have multiplicity at most three. We further show that a vertex-maximal 94 contains only triple points and “bad” fourfold points, and must contain a fourfold point all eight of whose angles are triangles (an *all-8 point*).

The proofs combine three ingredients:

- an exact budget identity, $\Lambda = 18 \cdot 16 - 3T = Z + \sum_P c_P$;
- a generalisation of the Bartholdi–Blanc–Loisel charging argument, made rigorous by per-line linear-programming certificates. These are checked exactly by dynamic programming over a finite automaton describing every possible line;
- local optimality lemmas, proved by exhaustive enumeration of local redrawings.

The same certificates hold at other n . They prove $K(14) = 54$ for *all* arrangements, a new exact value. For multiplicity at most three they prove the Bartholdi–Blanc–Loisel bound $\lfloor n(n - 7/3)/3 \rfloor$ for every even n from 6 to 40, which gives $K(16) = 72$ and $K(20) = 117$ in that class; previously these bounds were proved only for simple arrangements. For the remaining all-8 case we prove an exact identity $\Lambda = \pi + U + \frac{1}{2} \sum_P S_P^\circ + K_3 + 2K_4 + \frac{1}{2}\Delta$ with all terms nonnegative; $K(18) = 93$ is equivalent to $\Lambda \geq 7$ in this case, which is open. We release exact arrangements, triangle certificates, an independent checker, and all proof certificates with verification commands.

1 Introduction

Let $\mathcal{A}$ be an arrangement of n lines in the Euclidean plane and $T(\mathcal{A})$ the number of its bounded faces that are triangles, i.e. nondegenerate triangles not crossed by any line. The Kobon triangle problem, posed by Kobon Fujimura, asks for $K(n) = \max T(\mathcal{A})$.

The classical upper bound is Tamura’s $K(n) \leq \lfloor n(n - 2)/3 \rfloor$: each of the $n(n - 2)$ bounded segments of a simple arrangement borders at most two triangles, and the extremal configurations are restricted further. Clément and Bader [3] sharpened it by one for $n \equiv 0, 2 \pmod{6}$. Bartholdi, Blanc and Loisel [1] proved $T \leq \lfloor n(n - 7/3)/3 \rfloor$ for *simple* arrangements of lines and

*Computer-assisted work. The AI systems and compute that were used are disclosed in Section 11.

pseudolines; this is the source of the value 94 for $n = 18$ that is often quoted. Blanc [2] improved this to $\lfloor n(n - 5/2)/3 \rfloor$ for even n , which is 93 at $n = 18$. Both papers state and use simplicity.

For arrangements with points of multiplicity at least three, which occur in every known record for even n , we are not aware of a bound at $n = 18$ better than 96 (Tamura) or the unpublished 95 [3]. The best known construction has 93 triangles [4]. So, before this work, $K(18)$ was known only to lie in $\{93, 94, 95, 96\}$ (or $\{93, 94, 95\}$ if [3] is accepted).

Results. All results below concern arrangements of $n = 18$ pseudolines (each pair crossing exactly once); straight-line arrangements, including parallel lines, are covered by the standard projective treatment of parallel classes. “Computer-assisted” means that the proof reduces to exact computations whose certificates we publish.

Theorem 1 (computer-assisted). *Every arrangement of 18 pseudolines has at most 94 bounded triangles.*

Theorem 2 (computer-assisted). *Every arrangement of 18 pseudolines in which no point lies on four or more lines has at most 93 bounded triangles.*

Theorem 3 (computer-assisted). *Let $\mathcal{A}$ be an arrangement of 18 pseudolines with 94 bounded triangles, chosen with the maximum number of vertices. Then:*

- (a) *every multiple point of $\mathcal{A}$ is a triple point or a fourfold point whose cyclic word of eight sector types (triangle = 1) is, up to symmetry, 11111111, 11111110 or 11101110;*
- (b) *there is no point of type 11101110;*
- (c) *there is a point of type 11111111 (an all-8 point);*
- (d) *some fourfold point and some triple point are consecutive on a line, and every such consecutive pair has one of 48 explicitly listed local patterns;*
- (e) *every triple point has both alternating sums of its six sector bits at least 2.*

Theorems 1–3 reduce the question $K(18) = 93$ to excluding arrangements with an all-8 point. We do not complete this exclusion. Section 6 gives the sharpest reduction we have: an exact identity in nonnegative local credits. We also explain why the per-line method that proves Theorem 1 cannot finish the job as it stands.

On the construction side we found, by our own unseeded search, a new straight-line 18-line arrangement with 93 triangles and no multiple points. It is scored 93 by the competition’s exact evaluator, and 93 ties the record. It is *not* a new record.

Status. The proofs were developed with AI assistance (Section 11). They have been checked by exact computation, by independent reimplementations and by an independent audit, but not by a human referee or a proof assistant. Table 1 lists each claim with its evidence.

2 Preliminaries: the budget identity

Fix an arrangement $\mathcal{A}$ of n pseudolines. A *segment* is a bounded edge, i.e. a piece of a line between consecutive vertices. A segment is *used* by each triangle having it as a side, so it is used 0, 1 or 2 times. Let Z be the number of unused segments. For a point P of multiplicity $m_P \geq 3$ let:

- D_P be the number of rays at P whose first segment is used twice and ends at a simple vertex (*blocks*);
- β_P be the number of such rays ending at a multiple point (*bridges*);
- $N_P = 2m_P - D_P - \beta_P$ be the number of remaining rays.

Lemma 4 (budget identity). *With $\Lambda = n(n-2) - 3T$,*

$$\Lambda = Z + \sum_{m_P \geq 3} c_P, \quad c_P = m_P(m_P - 2) - D_P - \frac{1}{2}\beta_P.$$

For triple points $c_P = (N_P - D_P)/2$, and for fourfold points $c_P = 4 + (N_P - D_P)/2$. For $n = 18$, $\Lambda = 288 - 3T$ is divisible by 3; thus $T = 94 \iff \Lambda = 6$, and $T \leq 93 \iff \Lambda \geq 7$.

Proof sketch. Count triangle sides: $3T = (S - Z) + D_2$, where S is the number of segments and D_2 the number of segments used twice. A doubly used segment always has a multiple endpoint (L1): two triangles on both sides of a segment between two simple vertices would force a second crossing. Hence $D_2 = \sum_P D_P + \frac{1}{2} \sum_P \beta_P$. Finally, $S = n(n-2) - \sum_P m_P(m_P - 2)$, since each line through a point of multiplicity m loses $m - 2$ vertices. $\square$

Per-line form. Splitting $3c_P$ over the rays at P ($+\frac{3}{2}$ per N ray, $-\frac{3}{2}$ per block, 0 per bridge), and splitting each unused segment into three *portions*, gives

$$3\Lambda - n = \sum_L v_L + \text{waste}, \quad \text{waste} \geq 0.$$

One portion goes to its own line and one to the other line at each simple endpoint. v_L depends only on the line L and the faces incident to it. This is the per-line form of the Bartholdi–Blanc–Loisel argument [1]: in a simple arrangement with n even, every line has $v_L \geq 0$, which gives $\Lambda \geq n/3$. With multiple points individual lines can be negative, and charge must be moved between lines by *transfer rules*.

3 Per-line certificates

Transfer rules and the line automaton. We fix a finite catalogue of local *cells*: configurations around a block, a simple vertex, a triangle, or a multiple point that are visible from every line taking part. Each cell carries a transfer between the lines involved, with a weight w ; the transfers cancel exactly, so for every w

$$3\Lambda - 18 = \sum_L \text{final}_L(w) + \text{waste}.$$

To bound final_L from below for *every* line of every arrangement, we encode the view from a single line as a word over a finite alphabet of *frames*. A frame records the vertex type (simple, triple, fourfold), the triangle bits on both sides of the adjacent segments, and the multiplicity classes of triangle apexes. Hidden sector bits are also recorded, together with local facts that we prove separately. A line with 17 crossings is a path of exact weight 17 in a layered graph. final_L is a sum of window terms along the path, so $\min_L \text{final}_L$ is a shortest-path computation, i.e. a dynamic program (DP).

Certificates. A cutting-plane loop alternates between a linear program over the weights, the DP (which returns the worst path), and, where needed, SAT checks that a candidate worst path is not realisable. Proved unrealisable patterns are added to the automaton. The final weights are rationalised to a common denominator D , and the DP is rerun in exact integer arithmetic.

Proposition 5 (certificate FC). *For multiplicity ≤ 3 there are weights with $D = 16$ such that every automaton path has $\text{final} \geq 0$. Hence $\Lambda \geq 6$, i.e. $T \leq 94$.*

Proposition 6 (certificate FC-M). *For arbitrary multiplicity, with fourfold points restricted to the three bad words of Theorem 3(a) (Lemma 7) and the pair lemma (Lemma 8), there are weights with $D = 16$ such that every path has $\text{final} \geq -\frac{1}{4}$. Hence $3\Lambda - 18 \geq -18/4 > -9$. Since $3\Lambda - 18$ is a multiple of 9, $3\Lambda - 18 \geq 0$, i.e. $\Lambda \geq 6$.*

Propositions 5 and 6, with the reduction of Lemma 7, prove Theorem 1. The automaton facts used by the certificates were validated on several million real lines: no real line violates a fact, and the exact DP minimum is reproduced by an independent second DP.

4 Local optimality lemmas

If an arrangement with $T = 94$ exists, choose one with the maximum number of vertices. Redrawing a small disk around a point, with the same boundary crossings, gives another pseudoline arrangement. The redrawing can neither increase T (Theorem 1), nor keep T while increasing the number of vertices.

Lemma 7 (perturbation lemma). *In a vertex-maximal 94, every point of multiplicity ≥ 4 is a fourfold point with sector word 11111111, 11111110 or 11101110, up to symmetry.*

The local redrawings are enumerated exhaustively for multiplicities 4 and 5; for 6 they come from a sampled covering set; for $7 \leq m \leq 17$ there are explicit lossless simple redrawings. The enumeration was re-derived with an independent face computation by half-edge tracing (1,945 local words, 0 mismatches).

Lemma 8 (pair lemma). *In a vertex-maximal 94, if a fourfold point P and a triple point Q are consecutive on a line, then the pair of sector words of (P, Q) is one of 48 canonical patterns (out of 2,080). For an all-8 point P , the triple neighbour's word is one of 100111, 101101, 101111, 110011, 110111, 111111.*

Lemma 9 (triple optimality). *In a vertex-maximal 94, every triple point with sector bits $s_0, \dots, s_5$ satisfies $s_0 + s_2 + s_4 \geq 2$ and $s_1 + s_3 + s_5 \geq 2$.*

Proof. Resolving a triple point into three simple crossings creates one small triangle. It adds one side to each sector of one alternating triple, so $\Delta T = 1 - (s_0 + s_2 + s_4)$ or $1 - (s_1 + s_3 + s_5)$, while the vertex count grows by 2. $\square$

5 Excluding cases

Multiplicity at most three (Theorem 2). In a 94 we have $3\Lambda - 18 = 0$. So under every certificate with $\varepsilon = 0$, every line is *exactly* tight and the waste is zero. Intersecting the tight sets of five certificates (joint tightness) and iterating an exact elimination shows the following:

- there are no unused segments;
- only four point types survive;
- via a Gauss–Bonnet count on the bridge components, there are three structural cases: six isolated “X” points, one seven-point “flower” plus three X points, or two flowers.

The flower cases are closed by a tip lemma: two end conditions at opposite tips are jointly unsatisfiable among nine pseudolines, with a DRAT-certified SAT proof. The six-X case is closed by an independent cube SAT computation (2,841 cubes, all DRAT-verified). This chain was audited independently (five items, all passed).

Fourfold points. With fourfold points present, certificates with an extra *credit* at a fourfold point of a given word show $\sum_L \text{final}_L > 0$, contradicting $3\Lambda - 18 = 0$:

- word 11101110: margin $9/8$ (Theorem 3(b));
- word 11111110, when no all-8 point is present: margin $7/10$;
- a strict certificate shows some fourfold–triple adjacency is necessary (Theorem 3(d)).

Together with Theorem 2 and Lemma 7, a 94 must contain an all-8 point (Theorem 3(c)).

6 The remaining case: an all-8 point

Why the per-line method stops. For the all-8 case, the best per-line credit LP we could build has margin -2.29 instead of a positive one. This value is unchanged by every refinement we tried: hidden state along the line, further proved lemmas, the triple optimality filter, sixteen SAT-proved local patterns, and new kite-transfer rules. More decisively, SAT realises the binding paths as *actual* lines of arrangements in the class (29 of the 61 binding paths; 31 timed out and one was unrealisable). In those arrangements the line through the all-8 point is at the floor, while the other lines through the same point carry large surplus. Per-line accounting with local rules cannot move credit between the lines of a star.

A two-dimensional credit identity. Classify the blocks at their simple endpoint X : *touch* (U : the continuation beyond X is bounded and unused), *end* (I : the line ends at X), or *mutual*. Mutual blocks meet at a 3-fan or at a *kite* (a simple vertex with four triangles). Let K_q be the number of kites with exactly q fourfold corners.

Injective payments assign to every block a token: an N ray on a singly used segment, identified by its unique triangle. They give the following.

- No two consecutive blocks at a triple point, and no three at a fourfold point.
- Every kite with four triple corners has a singly used cap edge in each pair of opposite sides.
- Every kite with exactly one fourfold corner is paid. Either one of its cap edges is singly used, or its fourfold corner has slack two by a far-corner argument.
- Every touch or end block from a fourfold point receives a fresh token.

Consider the line ends next. Call an end of line L at the simple vertex $v = L \cap M$ a *bad wedge* if v is also the last vertex of M and the bounded face opposite the wedge is a triangle. Join two lines when they meet at a bad wedge.

Lemma 10 (forest lemma). *For n even, the bad-wedge graph is a forest of paths; let $\pi \geq 1$ be its number of components.*

Proof sketch. If both ends of L are bad wedges, the two corner triangles lie on the same side of L . A cycle of k lines then forces k odd, by counting crossings of the closed curve with one of its lines. Any line outside the cycle crosses the curve k times, which forces k even. So a cycle must contain every line, and n is odd. $\square$

Theorem 11 (credit identity). *For 18 pseudolines in the class of Theorem 3,*

$$\Lambda = \pi + U + \frac{1}{2} \sum_P S_P^\circ + K_3 + 2K_4 + \frac{1}{2}\Delta,$$

every term ≥ 0 . Here:

- $S_P^\circ = 8 - D_P - k_P - 2c_B(P)$ is the corrected slack at a fourfold point, where k_P counts its blocks ending at kites and $c_B(P)$ its kites with all cap edges doubly used;
- $\Delta \geq 0$ counts unused end slots and unused tokens.

So a 94 needs $\pi + U + K_3 + 2K_4 \leq 6$, i.e. at least 12 bad wedges, and $K(18) = 93$ is equivalent to

$$2\pi + 2U + \sum_P S_P^\circ + 2K_3 + 4K_4 + \Delta \geq 13$$

for this class. For simple arrangements, the forest lemma and the parity lemma below already give Blanc's case $\Lambda \geq 6$ in a few lines. Every line with both ends bad must contain an unused segment, so $\Lambda \geq \max(\pi, 18 - 2\pi) \geq 6$.

Lemma 12 (parity). *Let L have all bounded segments singly used and both ends bad wedges. Let t_L be the number of triple points on L , and f_L the number of touch or end centres capped by L . If n is even, $t_L + f_L$ is odd.*

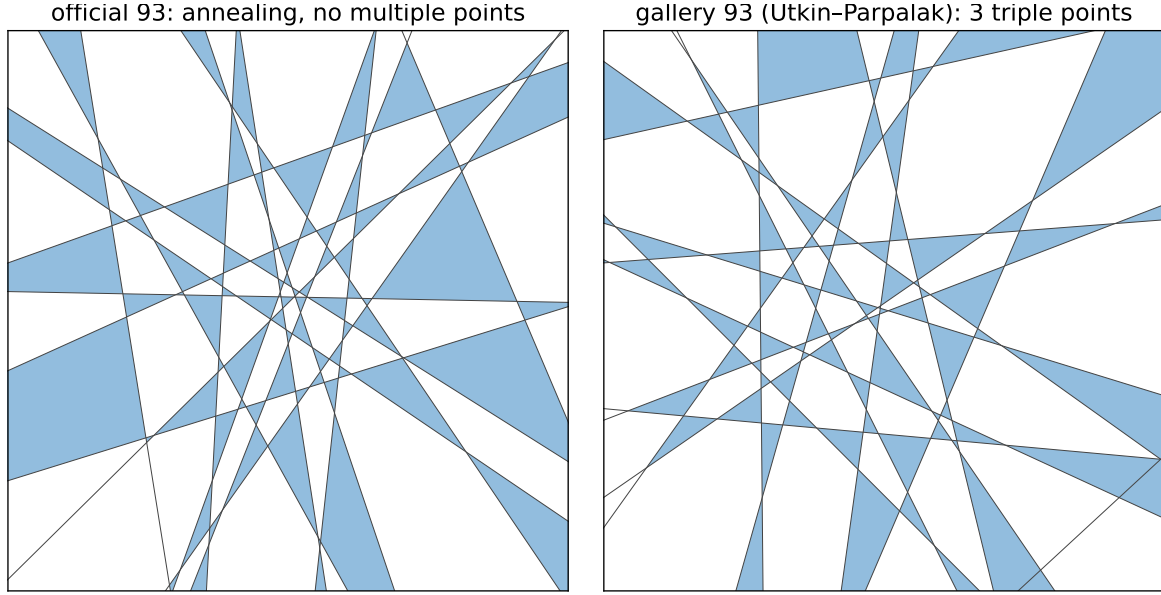


Figure 1: Two arrangements of 18 lines with 93 bounded triangles (shaded), central region. Some long thin triangles extend beyond the frame. Left: our arrangement (no multiple points), scored 93 by the exact evaluator. Right: an arrangement from [4] with three triple points. Exact coefficients and triangle lists are in `arrangements/`.

What is missing. The known 93s spread their $\Lambda = 9$ over all terms. Among 14,376 of them, $(\pi, U, \frac{1}{2}\Delta)$ ranges from $(9, 0, 0)$ to $(1, 0, 8)$ and $(2, 7, 0)$. So no single term can be forced. What is needed is a trade-off lemma in the class: many bad wedges force parity defects on the lines between them, and these defects force U or Δ . Lines through triple points flip parity at no cost, and this is the core difficulty. Partial results toward a two-stage finish: (B) $E \geq 18 - 3\pi$ would force a 94 into the zero-credit corner $\pi = 6$, $E = 0$. In that corner, hand counts of the lines forced by local saturation exclude every cluster of fourfold points except two local types. These are an isolated double-case-B star (at least 14 lines) and a two-point path with two case-B ends (at least 17 lines). The proof of (B) is reduced to a local capacity inequality that holds on all known data. Small cases are also decided by SAT: with an all-8 point in the class, $\Lambda \leq 6$ is impossible for $n \leq 10$, and the minimum is $\Lambda = 9$ at $n = 9$ and 8 at $n = 10$.

7 Constructions

Figure 1 shows two 18-line arrangements with 93 triangles:

- our arrangement, found by unseeded simulated annealing over line angles and offsets with a fast floating-point counter, then rounded to integer coefficients and recounted exactly. It has no multiple points;
- an arrangement with three triple points from the Utkin–Parnalak gallery [4].

Both are counted by the competition evaluator and by our independent checker (Section 9), with identical triangle sets. The frozen baseline of the competition, 18 tangents to a parabola, has 16 triangles. For $n = 15$ we also obtained the proven optimum 65 by the same search. Searches for a 94 found none:

- exact one-line extensions of all 255 perfect 17-line wiring diagrams;
- re-insertion of every line of 3,016 known 93s;
- 4-line merges of 42,331 arrangements with $T \geq 92$;
- SAT searches in many structured families.

8 Further results for general n

Theorem 13 (computer-assisted). *(a) For every even n with $6 \leq n \leq 40$, every arrangement of n pseudolines with no point on four or more lines has $T \leq \lfloor n(n - 7/3)/3 \rfloor$. This equals the known record at $n = 6, 8, 10, 12, 14, 16, 20$ [4, 5]. So in this class $K(n)$ is determined for these n , and in particular $K(16) = 72$ and $K(20) = 117$.*

(b) Every arrangement of 14 lines or pseudolines, of any multiplicity, has $T \leq 54$. Hence $K(14) = 54$.

Proof sketch. (a) Take the certificate FC of Proposition 5, with unchanged weights. Its ingredients are valid for all n : the automaton facts, the cell domains (an enumeration over automaton windows, filtered by local facts with hand proofs), and the base identity. Only the exact path weight $W = n - 1$ changes. Rebuilding the automaton for each even $n \leq 40$ and rerunning the exact DP gives $\min \text{final} = 0$. Hence $3\Lambda - n \geq 0$, and with $\Lambda \equiv n(n - 2) \pmod{3}$ the bound follows.

(b) The certificate FC-M of Proposition 6 verifies at $W = 13$ with $\min \text{final} \geq -\frac{1}{4}$. Its SAT-proved local patterns, which were proved for 18 lines, were re-proved for 14 lines with DRAT certificates. The perturbation and pair lemmas are local. Then $3\Lambda - 14 \geq -\frac{14}{4} > -5$. Since $\Lambda \equiv 0 \pmod{3}$ and $3 \cdot 3 - 14 = -5$, we get $\Lambda \geq 6$, i.e. $T \leq 54$. $\square$

The records at $n = 14$ and $n = 20$ use triple points and exceed Blanc’s bounds for simple arrangements (53 and 116), so the simple-arrangement theorems did not settle these values.

Two further results from the same framework hold for every even n . Their certificates and proofs are in the repository.

Theorem 14 (Theorem G). *Let n be even and let $\mathcal{A}$ be an arrangement of n lines or pseudolines (no three lines mutually parallel) with t triple points and no line through two multiple points. Then $\Lambda \geq n/2 - t$, i.e. $T \leq \lfloor (2n^2 - 5n + 2t)/6 \rfloor$. For $t = 0$ this is Blanc’s bound.*

Theorem 15 (automaton-certified, all even n). *For arrangements with only simple and triple points whose triple points belong to one of three restricted classes of local types, every line has $\text{final} \geq 0$ under explicit transfer rules. Hence $\Lambda \geq n/3$; for example $T \leq 54$ at $n = 14$ and $T \leq 72$ at $n = 16$.*

The second theorem is certified by an automaton without a length bound, so the certificate covers all n at once. Theorem G has a hand proof that was checked by an independent AI referee.

9 Verification and limitations

Independent checks.

- Triangle counts are verified by a standalone exact checker (rational arithmetic, two independent methods: half-edge face enumeration and a triple test against every other line).
- The perturbation and pair lemmas were re-derived with an independent half-edge face computation.
- The multiplicity- ≤ 3 chain was audited independently, with two independent encodings of the tip lemma.
- Every SAT exclusion used in a proof has a DRAT proof checked by `drat-trim`.

Limitations.

1. No human refereeing and no formalisation (no Lean or other proof assistant).

Claim	Evidence	Status
Thm. 1, mult. ≤ 3	exact certificate FC ($D = 16$), DP rerun exactly; second DP	computer-verified
Thm. 1, fourfold points	exact certificate FC-M ($D = 16, \varepsilon = \frac{1}{4}$)	computer-verified
Thm. 2	joint tightness, elimination, ILP; tip lemma (DRAT); 6-X case (2,841 DRAT cubes)	computer-verified, audited
Lemma 7, 8	exhaustive enumeration; independent re-derivation	computer-verified
Thm. 3(b), (d), and the 7-type case	exact credit certificates	computer-verified
Thm. 13(a)	certificate FC rechecked exactly at $W = n - 1$, even $n \leq 40$	computer-verified
Thm. 13(b), $K(14) = 54$	certificate FC-M at $W = 13$; patterns re-proved for 14 lines (DRAT)	computer-verified
Thm. 11 and lemmas of §6	hand proofs, each checked; exact checks on 3,474 arrangements	proofs checked, not refereed
$K(18) = 93$	—	open (all-8 case)

Table 1: Claims and evidence. “Computer-verified” means an exact rational or SAT/DRAT certificate that can be rechecked with the published scripts.

2. The automaton facts (“guarded facts”) used at fourfold apexes are validated on large real data sets, but their hand proofs are not all written out. (The two special-column identities used by the fourfold certificates, previously only validated on data, are now proved for every multiplicity; see the repository.)
3. The independent audit covers the multiplicity- ≤ 3 chain only, not the fourfold certificates.
4. SAT encodings are validated on many pinned cases but not formally verified.
5. The main open case is stated in Section 6.

10 Reproducibility

All files are in the accompanying repository:

- exact arrangements with triangle certificates: `arrangements/`;
- the independent checker: `checker/`;
- proof certificates with verification commands, expected outputs and logs: `certificates/`;
- the code and data they need: `search/`, `work/`, `tools/` (layout kept so that every command runs unchanged);
- technical notes, including the full proofs of the lemmas in Section 6: `proofs/`;
- a catalogue of 14,376 distinct 93-triangle 18-pseudoline arrangements (wiring words, 0 to 21 triple points): `data/`.

Dependencies are pinned in `requirements.txt`; `REPRODUCE.md` lists every command.

11 Disclosure

This work was carried out with AI systems:

- Claude Opus 5.5 (Anthropic), in Claude Code, did the planning, the mathematics and the checking of all proofs;
- Claude Sonnet 5.5 subagents wrote software and performed audits;
- GPT 6.1 contributed the notes behind the credit identity of Section 6 (forest lemma, injective payments). Every step was checked before use.

Computation ran on one 24-core workstation (62 GB RAM). The construction was scored on the AutoLab platform (competition hill `alejandrozu/kobon-triangles`, official report 203bd68f, 93 triangles at $n = 18$). The author directed the work and takes responsibility for the claims.

References

- [1] N. Bartholdi, J. Blanc, S. Loisel, *On simple arrangements of lines and pseudo-lines in $\mathbb{P}^2$ and $\mathbb{R}^2$ with the maximum number of triangles*, arXiv:0706.0723 (2007).
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- [4] A. Utkin, R. Parpalak, online gallery of Kobon triangle arrangements, <https://parpalak.com> (accessed September 2026).
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